Assignment 01



6 BAR TOGGLE MECHANISM

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SECTION 1.1

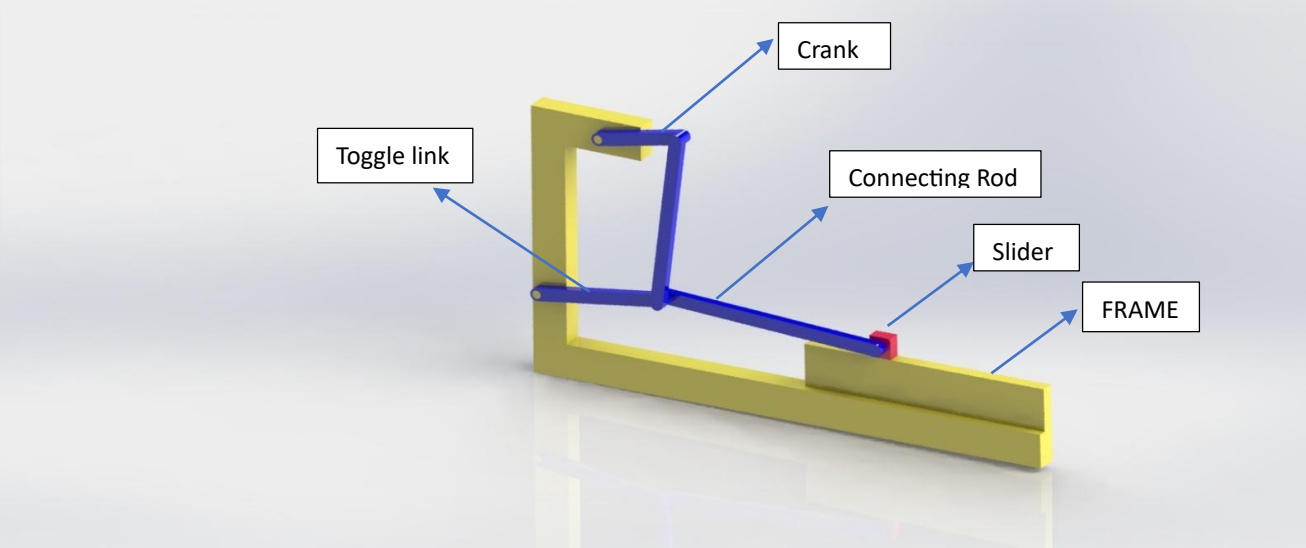


Fig.1 CAD model of toggle mechanism prepared in Solidworks

SECTION 1.2

Description of the Six-Bar Toggle Mechanism

A toggle mechanism is a specific type of mechanical linkage that utilizes two or more links to create a significant force amplification based on the input movement. In the context of a six-bar toggle mechanism, it consists of six links and seven joints, allowing for complex motion and force transformation.

Structure and Components

- **Links**: The six-bar toggle mechanism includes six distinct links. These can be categorized into:
  - **Crank**: The link where the input speed in RPM is applied.
  - **Slider**: The link that delivers the amplified output force.
  - **Connecting Links**: The intermediate links that transmit motion and force between the input and output links.
- **Joints**: There are seven joints connecting these links, typically using pin (hinge) connections that allow rotation.

Types of Six-Bar Linkages

The six-bar mechanism can be configured in different topologies, primarily:

- **Watt Topology**: This configuration includes two four-bar loops, which helps in generating parallel motion and can trace nearly straight-line paths.

- **Stephenson Topology**: This configuration features one four-bar loop and one five-bar loop, where two ternary links are connected by a binary link. This setup allows for unique transmission characteristics, making it suitable for applications requiring high mechanical advantage

Working Principle

The toggle mechanism operates on the principle that a small input force can produce a much larger output force. This is achieved through the arrangement of the links and their angles during operation:

- When the input link is rotated, it causes other links to move. As these links approach a straight line (180 degrees), the output force increases dramatically due to the mechanical advantage offered by the geometry of the linkage.
- For instance, in applications like rock crushing or metal forming, a small force applied at one end can result in an output force that is significantly larger at another point, often achieving ratios of up to 29 times the input force depending on the angles involved

SECTION 1.3

Application

Different form of Toggle mechanism are used as follow:

1. Press Machines

- Used in sheet metal stamping, forging, and punching operations.

2. Injection Moulding Machines

- Helps in closing and locking the mould during plastic injection.

3. Rock Crushers

- Used to apply large crushing forces for breaking stones or ores.

4. Clamping Devices

- Used in fixture and jig mechanisms to hold work pieces securely.

5. Mechanical Press Brakes

- Used for bending sheet metal in automotive and manufacturing industries.
- The toggle linkage ensures precise control of the bending force.

6. Toggle Clamps

- Used in industrial automation, woodworking, and welding fixtures.

7. Automobile Lifting Jacks

- Some scissor jacks use a toggle mechanism for efficient lifting with minimal manual effort.

The toggle six-bar mechanism is widely used where force multiplication and mechanical advantage are required, making it essential in manufacturing, automotive, and automation industries.

SECTION 2.1

Pictorial Depiction

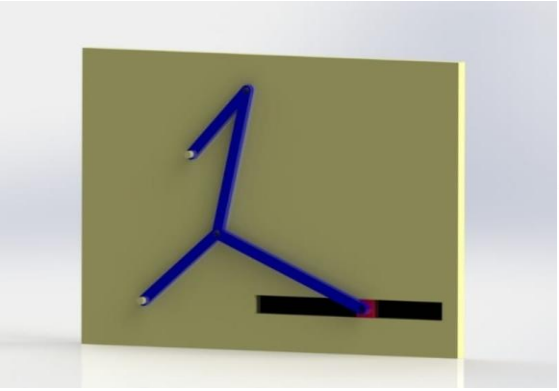


Fig. 1.1 Simulated Cad model

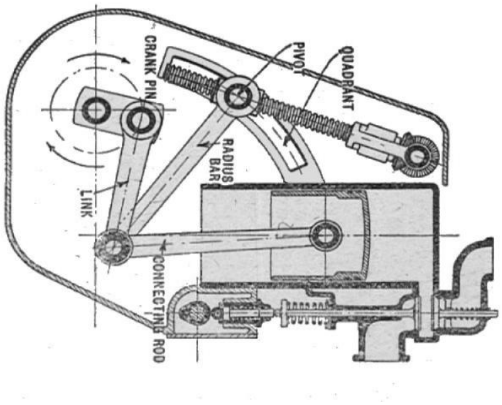


Fig. 1.2 Toggle mechanism as variable stoke engine

Above seen cad model is made on solidworks and exported to Adams for kinematics simulation. As mentioned earlier Toggle mechanism is used in different machine in different forms among them one is as variable stokes engine which is shown in the above figure 1.2.

SECTION 2.2

Geometry and Material:

Section 2.1.1

Material-Steel

Link(mm)	1	2	3	4	slider
Length	200	400	300	500	50
Width	30	30	30	30	30
Thickness	20	20	20	20	20

Section 2.1.1

Mass:

Link	1	2	3	4	slider
Mass(kg)	1.0464040393	2.0315391679	1.4164337821	2.4750916036	0.5115523071

Section 2.1.3

Moment of Inertia:

Link	1	2	3	4	slider
$I_{xx}(m^4)$	4.356304E-03	3.1359189E-02	1.1130506E-02	5.7345104E-02	2.4010511E-04
$I_{yy}(m^4)$	4.331592E-03	3.1355133E-02	1.1068654E-02	5.7289573E-02	1.5841898E-04
$I_{zz}(m^4)$	1.281125E-04	2.9974906E-04	1.5628068E-04	3.0914700E-04	1.5841898E-04

SECTION 2.3

Constraints:

Joints	Parts involved	Type of kinematic pairs
1	Link 1 and ground	Revolute
2	Link 3 and ground	Revolute
3	Link 2 and link 3	Revolute
4	Link 4 and link 3	Revolute
5	Link 1 and link 2	Revolute
6	Slider and link 4	Revolute
7	Ground and slider	Translation

SECTION 2.4

Motion Transmission:

Driving Component: link 1 is my driving component which is crank rotating at 210 RPM as input. Which can be rotated by motor in machine when implemented in real life machine.

Output link: slider is my output which gives me required output according to my input. Which can be used for cutting, pushing, crushing etc. when implemented in real life machines.

Kinematic simulation

In the Kinematic simulation, we analyse how the output component moves—its displacement, velocity, and acceleration—by varying the angular velocity at the crank over a set period. Then, we test the mechanism's sensitivity by introducing clearance to one joint and running it at standard and higher angular velocities. Analytical methods help us understand the component's motion characteristics. Finally, we represent all the data graphically, clearly understanding how the component behaves under different conditions. This comprehensive approach aids in assessing and interpreting the dynamic performance of the element.

SECTION 3.1

Output component:

We're examining how the piston responds to a specific angular velocity at the crank. The analysis includes tracking displacement, velocity, and acceleration. We've set a constant angular velocity of 210 rpm at Joint 1. The simulation is run for 5 seconds.

Section 3.1.1

The simulated Displacement profile of the output component is plotted in the graph below:

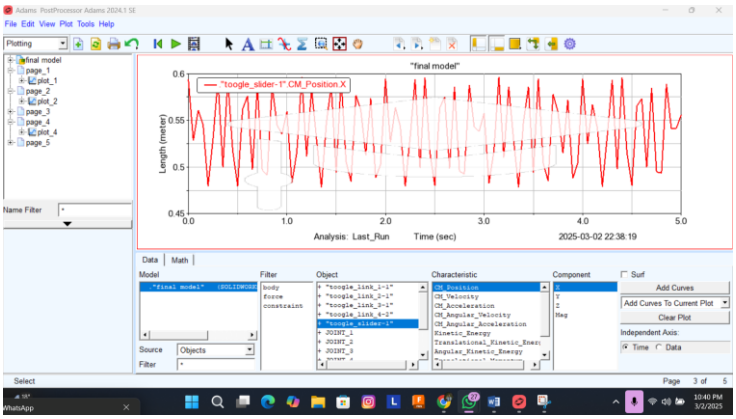


Fig. 3.1 Displacement profile

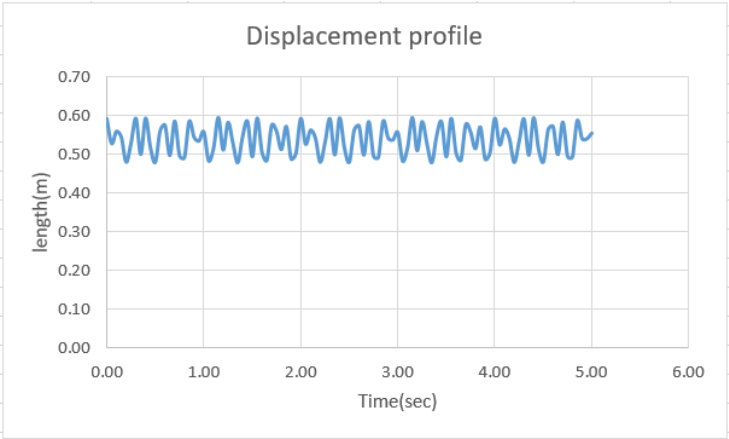


Fig. 3.2 Displacement profile from excel

Section 3.1.2

The simulated Velocity profile of the output component is plotted in the graph below:

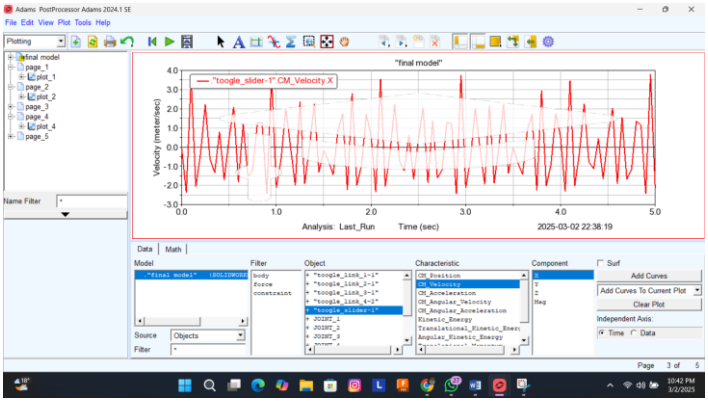


Fig. 3.3 Velocity profile

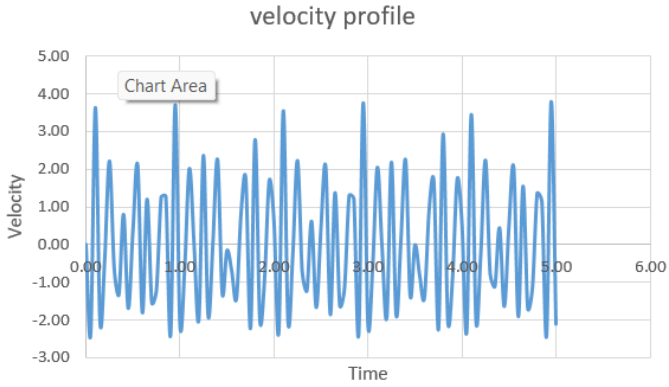


Fig. 3.4 Velocity profile from excel

Section 3.1.3

The simulated Acceleration profile of the output component is plotted in the graph below:

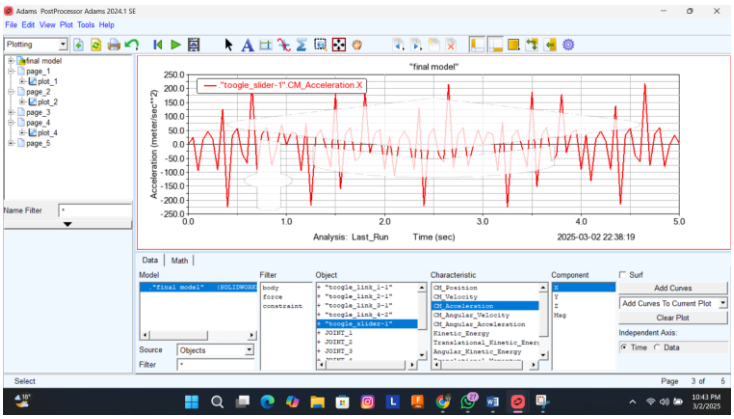


Fig. 3.5 Acceleration profile

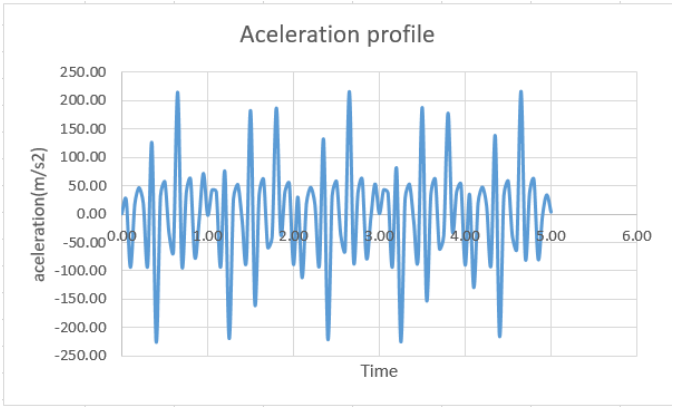


Fig. 3.6 Acceleration profile from excel

SECTION 3.2

Sensitivity Analysis

Link(mm)	1	2	3	4	slider
Length	200	400	305	505	50
Width	30	30	30	30	30
Thickness	20	20	20	20	20

I have increased the length of link 3 and 4 by 5mm.

The various plots and their comparisons before and after change of length are shown below:

1. Displacement curve:

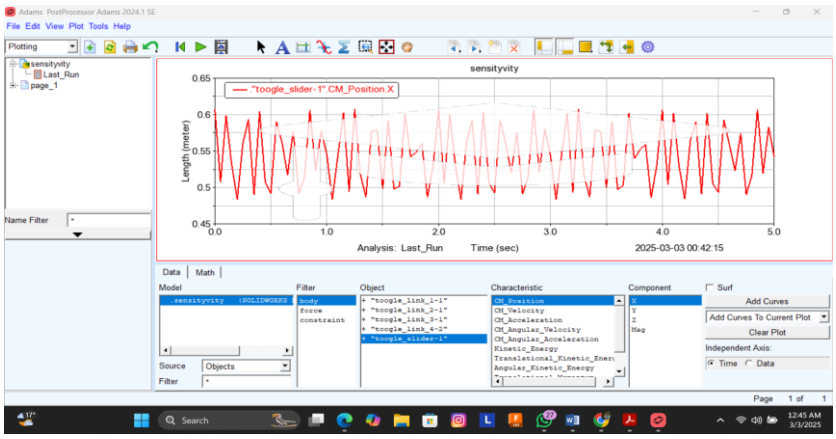


Fig.3.7 Displacement profile

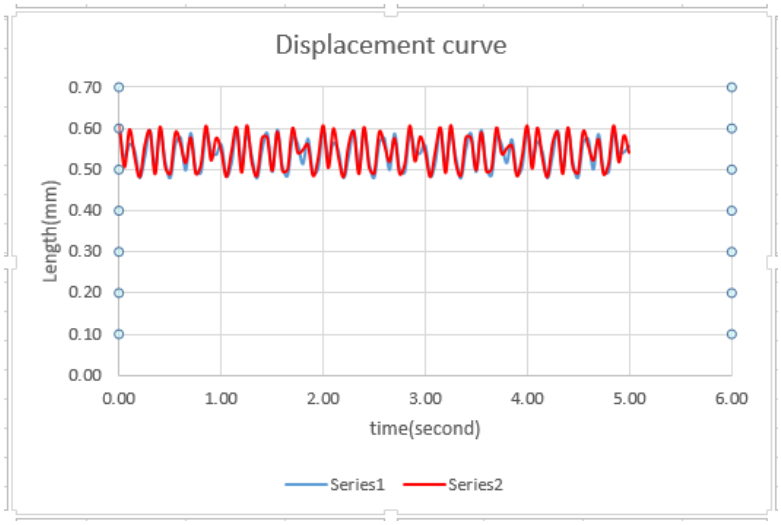


Fig.3.8 Displacement profile from excel

In displacement there is minor change before and after changing of length.

Velocity curve

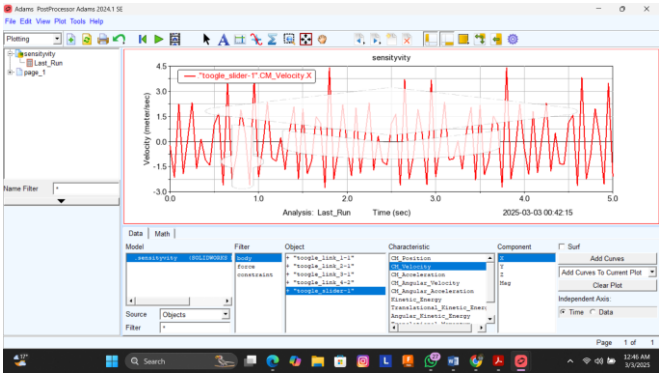


Fig. 3.9 velocity profile

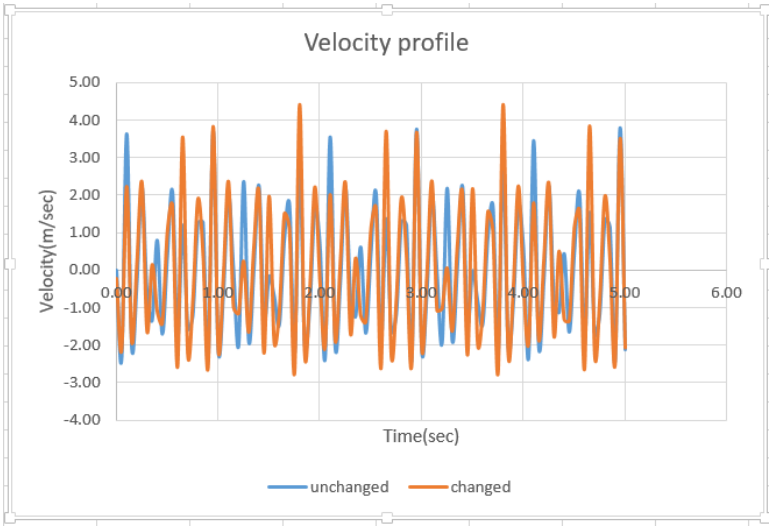


Fig. 3.10 velocity profile from excel

In velocity the changes are seen clearly before and after the changes are made.

Acceleration Curve:

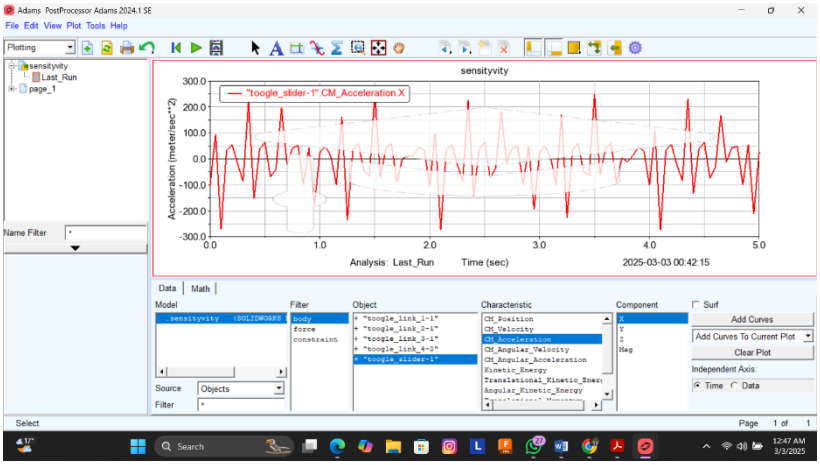


Fig. 3.11 Acceleration profile

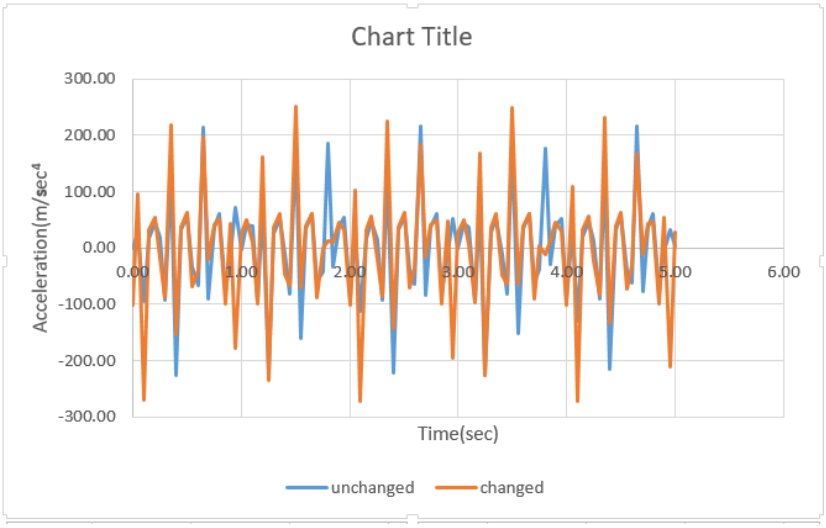


Fig. 3.13 Acceleration profile from excel

Significant Changes are clearly visible in acceleration also.

SECTION 3.3

Clearance

I have introduced clearance of 1mm at the joint of link 1 and link 2.

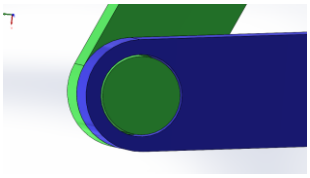


Fig. 3.14 clear view of applied clearance

Displacement curve after clearance when angular velocity is normal (22rad/sec):

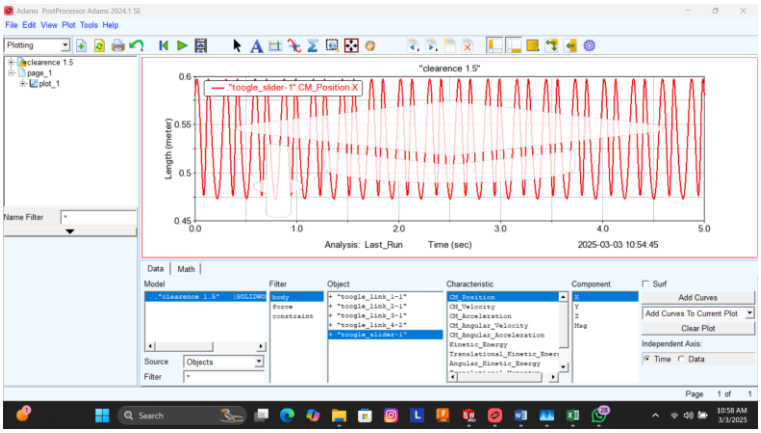


Fig. 3.15 Displacement profile

Displacement curve after clearance when angular velocity is high (25rad/sec):

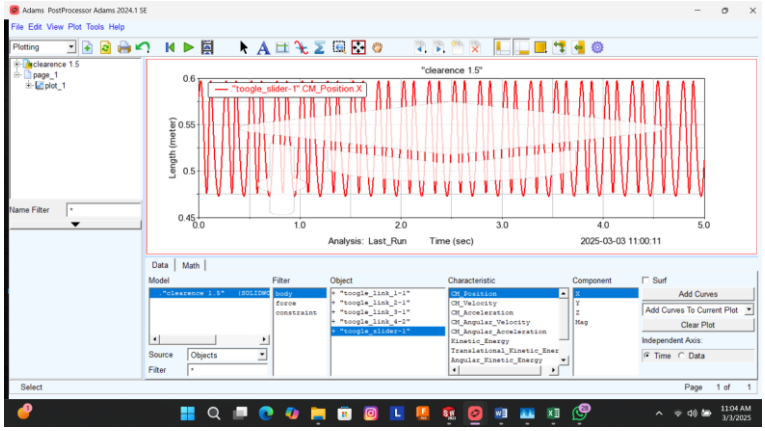


Fig. 3.16 Displacement profile from excel

Two displacement curve with clearance And Normal mechanism displacement curve:

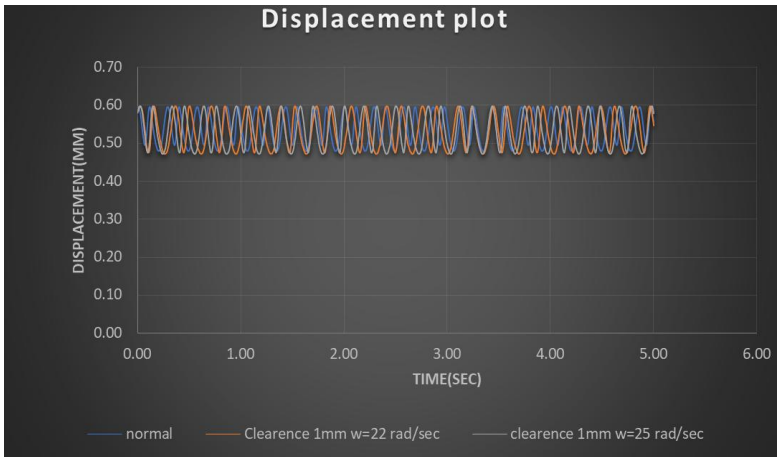


Fig. 3.17 Displacement profile from excel

Observation

After introducing the clearance the change in displacement curve is not Cleary visible for 100 steps due to error so I have decreased the error by simulating up to 10000 steps and then the variation is clearly seen between three cases.

SECTION 4

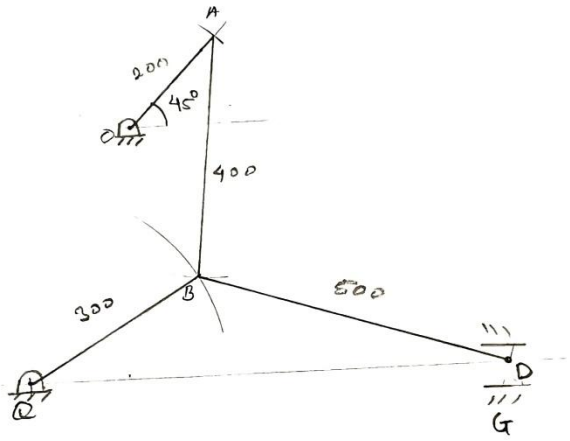
Validation of kinematic simulation:

Section 4.1.1

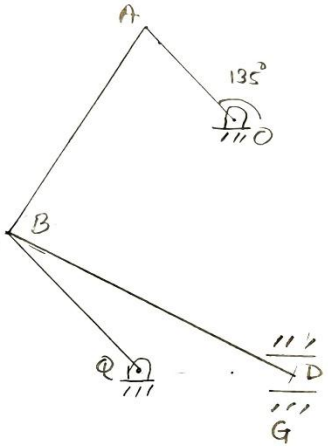
Displacement analysis by graphical method

As the displacement analysis the position of mechanism in four different position as shown with respect to 4 position of crank that is at 45°, 135°, 225° and 315° :

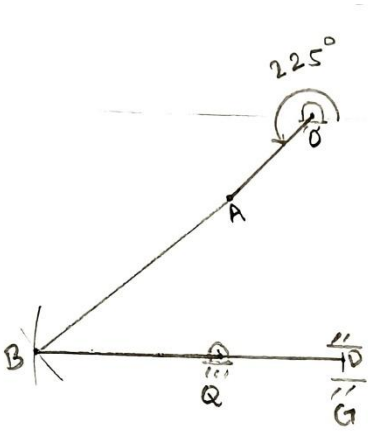
When θ=45°



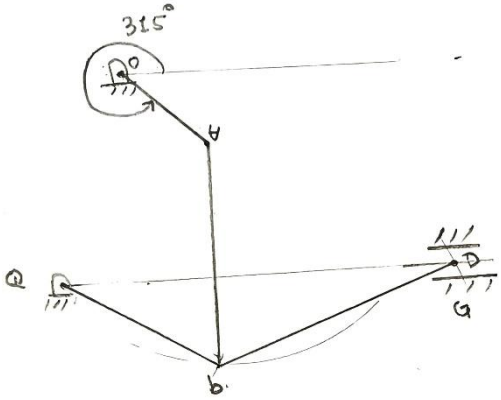
When θ=135°



When $\boldsymbol{\theta}$=225°



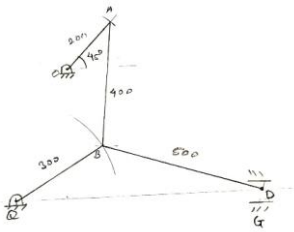
When $\boldsymbol{\theta}$=315°



Section 4.1.2

Velocity analysis by graphical method:

When θ=45°



N=210 RPM

$$\omega_{oA} = 22\ rad/s$$

$$V_{oA} = \omega_{0A} r_{0A}$$

$$V_{oA} = 22 \times 0.2$$

$$V_{oA} = 4.4\ m/s$$

Take the scale for the velocity diagram to be $50mm = 4.4\ m/s$.

$$V_{oA} = 4.4\ m/s$$

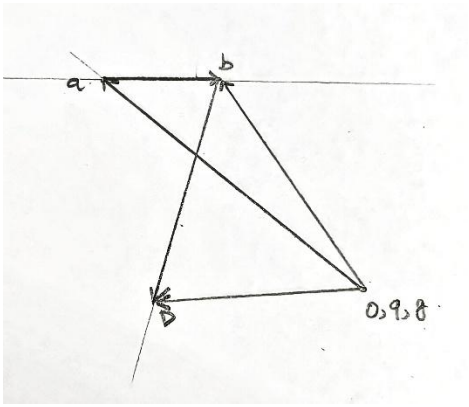


Fig. velocity polygon

$$V_{OA} = 4.4m/s$$

$$V_{AB} = 1.54\ m/s$$

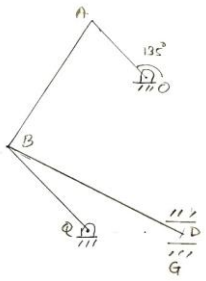
$$V_{BQ} = 3.344\ m/s$$

$$V_{DB} = 3.08\ m/s$$

$$V_{DG} = 2.72\ m/s$$

The velocity of the slider is $V_{DG} = 2.72m/s$

When θ=135°



N=210 RPM

$$\omega_{oA} = 22\ rad/s$$

$$V_{oA} = \omega_{0A} r_{0A}$$

$$V_{oA} = 22 \times 0.2$$

$$V_{oA} = 4.4\ m/s$$

Take the scale for the velocity diagram to be $50mm = 4.4\ m/s$.

$$V_{oA} = 4.4\ m/s$$

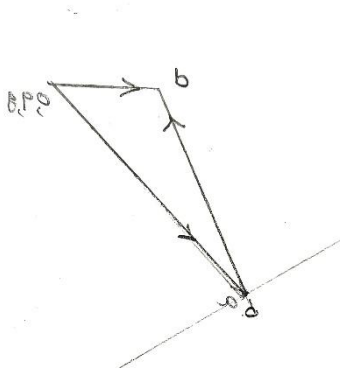


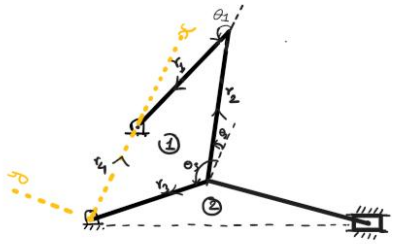
Fig. velocity polygon

$$V_{OA} = 4.4m/s$$
$$V_{AB} = 0.088\ m/s$$
$$V_{BQ} = 4.488\ m/s$$
$$V_{DB} = 3.432\ m/s$$
$$V_{DG} = 1.672\ m/s$$

The velocity of the slider is $V_{DG} = 1.672m/s$

Section 4.1.3

Velocity analysis by analytical method:



Loop closure equations in loop 1.

$$r_1\cos\theta_1 + r_3\cos\theta_3 = r_2\cos\theta_2 + r_4 \quad \text{——— (1)}$$

$$r_1\sin\theta_1 + r_3\sin\theta_3 = r_2\sin\theta_2 \quad \text{——— (2)}$$

differentiating

$$r_1\dot{\theta}_1\sin\theta_1 + r_3\dot{\theta}_3\sin\theta_3 = r_2\dot{\theta}_2\sin\theta_2 \quad \text{——— (3)}$$

$$r_1\dot{\theta}_1\cos\theta_1 + r_3\dot{\theta}_3\cos\theta_3 = r_2\dot{\theta}_2\cos\theta_2 \quad \text{——— (4)}$$

multiplying eqn (3) with $\cos\theta_2$ (4) with $\sin\theta_2$ and substract.

$$r_1\dot{\theta}_1\sin\theta_1\cos\theta_2 + r_3\dot{\theta}_3\sin\theta_3\cos\theta_2 = r_2\dot{\theta}_2\sin\theta_2\cos\theta_2$$

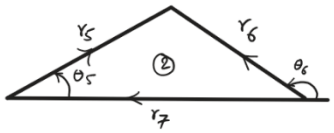
$$r_1\dot{\theta}_1\cos\theta_1\sin\theta_2 + r_3\dot{\theta}_3\sin\theta_2\cos\theta_3 = r_2\dot{\theta}_2\sin\theta_2\cos\theta_2$$

$$r_1\dot{\theta}_1(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2) = -r_3\dot{\theta}_3(\sin\theta_3\cos\theta_2 - \sin\theta_2\cos\theta_3)$$

$$r_1\dot{\theta}_1\sin(\theta_1-\theta_2) = r_3\dot{\theta}_3\sin(\theta_2-\theta_3)$$

$$\boxed{\dot{\theta}_3 = \frac{r_1\dot{\theta}_1\sin(\theta_1-\theta_2)}{r_3\sin(\theta_2-\theta_3)}}$$ angular velocity of link crank (3)

Loop 2



$$r_7 + r_5\cos\theta_5 = r_6\cos\theta_6$$

$$r_6\cos\theta_6 - r_5\cos\theta_5 = r_7 \quad \text{——— (i)}$$

$$r_6\sin\theta_6 - r_5\sin\theta_5 = 0 \quad \text{——— (ii)}$$

diff w.r.t time,

$$-r_6\dot{\theta}_6\sin\theta_6 + r_5\dot{\theta}_5\sin\theta_5 = \dot{r}_7 \quad \text{(iii)}$$

$$r_6\dot{\theta}_6\cos\theta_6 - r_5\dot{\theta}_5\cos\theta_5 = 0 \quad \text{(iv)}$$

multiplying $\cos\theta_6$ with (iii) and $\sin\theta_6$ with (iv) and adding.

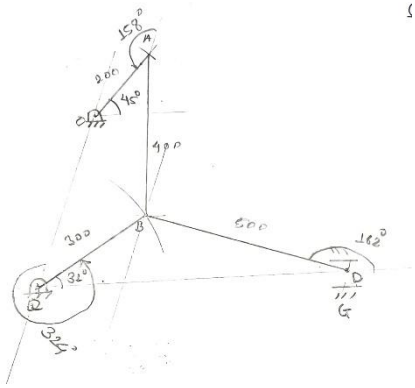
$$-r_6\dot{\theta}_6\sin\theta_6\cos\theta_6 + r_5\dot{\theta}_5\sin\theta_5\cos\theta_6 = \dot{r}_7\cos\theta_6$$

$$r_6\dot{\theta}_6\cos\theta_6\sin\theta_6 - r_5\dot{\theta}_5\cos\theta_5\sin\theta_6 = 0$$

$$r_5\dot{\theta}_5\sin(\theta_5-\theta_6) = \dot{r}_7\cos\theta_6$$

$$\dot{r}_7 = \frac{r_5\dot{\theta}_5\sin(\theta_5-\theta_6)}{\cos\theta_6}$$

Calculation of velocity for position 1($\theta = 45°$)



for position ①

$$\dot{\theta}_3 = \frac{r_1 \dot{\theta}_1 \sin(\theta_1 - \theta_2)}{r_3 \sin(\theta_2 - \theta_3)}$$

$$\dot{\theta}_3 = \frac{0.2 \times 22 \sin(158 - 16)}{0.3 \sin(16 - 324)}$$

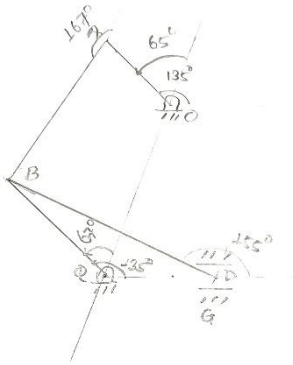
$$= 11.45$$

$$\dot{r}_7 = \frac{r_5 \dot{\theta}_5 \sin(\theta_5 - \theta_6)}{\cos\theta_6}$$

$$\dot{r}_7 = \frac{0.3 \times 11.45 \sin(162 - 31)}{\cos 162}$$

$$\dot{r}_7 = 2.725 \text{ m/sec}$$

Calculation of velocity for position 2($\theta = 135°$)



for position ②

$$\dot{\theta}_3 = \frac{r_1 \dot{\theta}_1 \sin(\theta_1 - \theta_2)}{r_3 \sin(\theta_2 - \theta_3)}$$

$$\frac{0.2 \times 22 \sin(65° - 167°)}{0.3 \sin(167° - 67°)}$$

$$-14.56 \text{ rad/sec}$$

$$\dot{r}_7 = \frac{r_5 \dot{\theta}_5 \sin(\theta_5 - \theta_6)}{\cos\theta_6}$$

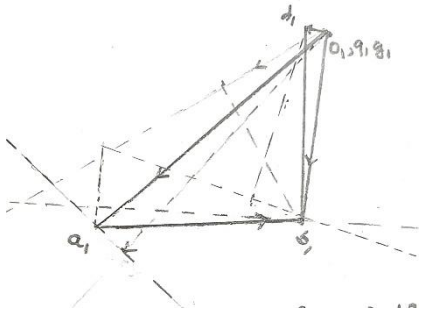
$$\dot{r}_7 = \frac{-0.3 \times 14.56 \sin(155 - 135)}{\cos 155°}$$

$$\dot{r}_7 = 1.65 \text{ m/sec}$$

Section 4.1.4

Acceleration analysis by graphical method:

When θ=45°



$$\vec{a}_{OA} = (\vec{a}_{OA})_r = \frac{V_{OA}^2}{r_{OA}} = 96.8\ m/s^2$$

$$(\vec{a}_{OA})_t = \alpha r = 12\ m/s^2$$

Assuming angular acceleration of 60 rad/sec

Here $(\vec{a}_{BA})_r$ is the radial component of acceleration in Link OA.

Here $(\vec{a}_{BA})_t$ is the tangential component of acceleration in Link OA.

Take the scale for the acceleration diagram to be $50mm = 96.8m/s^2$.

$$(\vec{a}_{AB})_r = \frac{V_{AB}^2}{r_{AB}} = \frac{1.54^2}{0.4} = 5.92\ m/s^2$$

$$(\vec{a}_{AB})_r = 5.92\ m/s^2$$

$$(\vec{a}_{BQ})_r = \frac{V_{BQ}^2}{r_{BQ}} = \frac{3.344^2}{0.3} = 37.27\ m/s^2$$

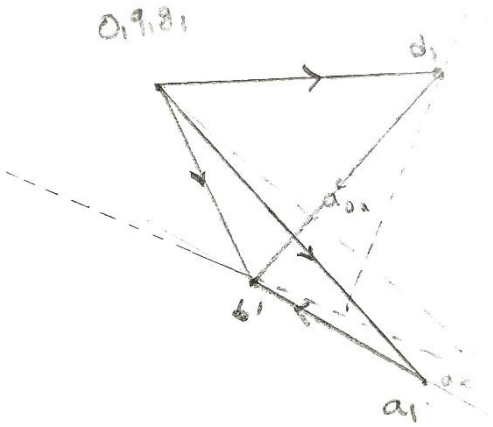
$$(\vec{a}_{BQ})_r = 37.27 m/s^2$$

$$(\vec{a}_{BD})_r = \frac{V_{BD}^2}{r_{BD}} = \frac{3.08^2}{0.5} = 18.97 m/s^2$$

$$(\vec{a}_{BD})_r = 18.97 m/s^2$$

So, the value of the acceleration of the slider is: $9.86\ m/s^2$ by measuring

When θ=135°



$$\vec{a}_{OA} = (\vec{a}_{OA})_r = \frac{V_{OA}^2}{r_{OA}} = 96.8\ m/s^2$$
$$(\vec{a}_{OA})_t = \alpha r = 12\ m/s^2$$

Assuming angular acceleration of 60 rad/sec
Here $(\vec{a}_{BA})_r$ is the radial component of acceleration in Link OA.
Here $(\vec{a}_{BA})_t$ is the tangential component of acceleration in Link OA.
Take the scale for the acceleration diagram to be $50mm = 96.8m/s^2$.

$$(\vec{a}_{AB})_r = \frac{V_{AB}^2}{r_{AB}} = \frac{0.088^2}{0.4} = 0.01936\ m/s^2$$
$$(\vec{a}_{AB})_r = 0.01936\ m/s^2$$

$$(\vec{a}_{BQ})_r = \frac{V_{BQ}^2}{r_{BQ}} = \frac{4.488^2}{0.3} = 50.35\ m/s^2$$
$$(\vec{a}_{BQ})_r = 50.35\ m/s^2$$

$$(\vec{a}_{BD})_r = \frac{V_{BD}^2}{r_{BD}} = \frac{3.432^2}{0.5} = 50.35 m/s^2$$
$$(\vec{a}_{BD})_r = 50.35 m/s^2$$

So, the value of the acceleration of the slider is: $18.06\ m/s^2$ by measuring

Section 4.1.5

Acceleration analysis by analytical method:

again diff. to get acceleration

$$r_1\ddot{\theta}_1\sin\theta_1 + r_1\dot{\theta}_1\cos\theta_1 + r_3\ddot{\theta}_3\sin\theta_3 + r_3\dot{\theta}_3\cos\theta_3 = r_2\ddot{\theta}_2\sin\theta_2 + r_2\dot{\theta}_2\cos\theta_2 \quad \text{—⑤}$$

$$r_1\ddot{\theta}_1\cos\theta_1 - r_1\dot{\theta}_1\sin\theta_1 + r_3\ddot{\theta}_3\cos\theta_3 - r_3\dot{\theta}_3\sin\theta_3 = r_2\ddot{\theta}_2\cos\theta_2 - r_2\dot{\theta}_2\sin\theta_2 \quad \text{—⑥}$$

Two eleminate one
multiply ⑤ with $\cos\theta_2$ and ⑥ with $\sin\theta_2$ and subtract.

$$r_1\ddot{\theta}_1\sin\theta_1\cos\theta_2 + r_1\dot{\theta}_1\cos\theta_1\cos\theta_2 + r_3\ddot{\theta}_3\sin\theta_3\cos\theta_2 + r_3\dot{\theta}_3\cos\theta_3\cos\theta_2 = r_2\ddot{\theta}_2\sin\theta_2\cos\theta_2 + r_2\dot{\theta}_2^2\cos^2\theta_2$$

$$r_1\ddot{\theta}_1\cos\theta_1\sin\theta_2 - r_1\dot{\theta}_1\sin\theta_1\sin\theta_2 + r_3\ddot{\theta}_3\cos\theta_3\sin\theta_2 - r_3\dot{\theta}_3\sin\theta_3\sin\theta_2 = r_2\ddot{\theta}_2\cos\theta_2\sin\theta_2 - r_2\dot{\theta}_2^2\sin^2\theta_2$$

$$r_1\ddot{\theta}_1\sin(\theta_1-\theta_2) + r_1\dot{\theta}_1^2\cos(\theta_1-\theta_2) + r_3\ddot{\theta}_3\sin(\theta_3-\theta_2) = r_2\dot{\theta}_2^2(\cos^2\theta_2+\sin^2\theta_2)$$

$$r_3\ddot{\theta}_3\sin(\theta_3-\theta_2) = r_2\dot{\theta}_2^2 - r_1\ddot{\theta}_1\sin(\theta_1-\theta_2) - r_1\dot{\theta}_1^2\cos(\theta_1-\theta_2)$$

$$\boxed{\ddot{\theta}_3 = \frac{r_2\dot{\theta}_2^2 - r_1\ddot{\theta}_1\sin(\theta_1-\theta_2) - r_1\dot{\theta}_1^2\cos(\theta_1-\theta_2)}{r_3\sin(\theta_3-\theta_2)}}$$

---

diff again,

$$-r_6\dot{\theta}_6^2\cos\theta_6 - r_6\ddot{\theta}_6\sin\theta_6 + r_5\dot{\theta}_5^2\cos\theta_5 + r_5\ddot{\theta}_5\sin\theta_5 = \ddot{r}_7 \quad \text{(v)}$$

$$-r_6\ddot{\theta}_6^2\sin\theta_6 + r_6\ddot{\theta}_6\cos\theta_6 + r_5\ddot{\theta}_5^2\sin\theta_5 - r_5\ddot{\theta}_5\cos\theta_5 = 0 \quad \text{(vi)}$$

multiply v with $\cos\theta_6$ and vi with $\sin\theta_6$ and

$$-r_6\dot{\theta}_6^2\cos^2\theta_6 - r_6\ddot{\theta}_6\sin\theta_6\cos\theta_6 + r_5\dot{\theta}_5^2\cos\theta_5\cos\theta_6 + r_5\ddot{\theta}_5\sin\theta_5\cos\theta_6 = \ddot{r}_7\cos\theta_6$$

$$-r_6\dot{\theta}_6^2\sin^2\theta_6 + r_6\ddot{\theta}_6\sin\theta_6\cos\theta_6 + r_5\dot{\theta}_5^2\sin\theta_5\sin\theta_6 - r_5\ddot{\theta}_5\cos\theta_5\sin\theta_6 = 0$$

$$-r_6\dot{\theta}_6^2 + r_5\dot{\theta}_5^2\cos(\theta_5-\theta_6) + r_5\ddot{\theta}_5\sin(\theta_5-\theta_6) = \ddot{r}_7\cos\theta_6$$

$$\ddot{r}_7 = \frac{-r_6\dot{\theta}_6^2 + r_5\dot{\theta}_5^2\cos(\theta_5-\theta_6) + r_5\ddot{\theta}_5\sin(\theta_5-\theta_6)}{\cos\theta_6}$$

where $\dot{\theta}_6 = \frac{r_5\dot{\theta}_5\cos\theta_5}{r_6\cos\theta_6}$

Acceleration of slider

Calculation of acceleration for position 1($\theta = 45°$)

for $\theta = 45°$.

$\dot{\theta}_6 \Rightarrow -6.19$

$\dot{\theta}_5 \Rightarrow 11.45$

$\ddot{\theta}_5 \Rightarrow \ddot{\theta}_3 = \dfrac{r_2\dot{\theta}_2^2 - r_1\ddot{\theta}_1 \sin(\theta_1-\theta_2) - r_1\dot{\theta}_1^2\cos(\theta_1-\theta_2)}{r_3\sin(\theta_3-\theta_2)}$

$\Rightarrow \dfrac{0.4\times(-3.36)^2 - 0.2\times 22^2\cos(158-16^\circ)}{0.3\sin(324-16^\circ)}$

$\Rightarrow -341.76.$

$$\ddot{r}_7 = -0.5(-6.19)^2 + 0.3\times(11.45)^2\cos(31-162) + 0.3\times(-341.76)\sin(31-162)/\cos 162$$

$\Rightarrow$ 11.32 m/Sec^2.

Calculation of acceleration for position 2($\theta = 135°$)

for $\theta = 135°$.

$\ddot{\theta}_6 = \dfrac{0.3\times 14.56\cos 135^\circ}{0.5\cos(155^\circ)}$

$\Rightarrow 6.81$

$\dot{\theta}_3 = \dot{\theta}_5 = 14.56$ (from velocity analysis)

$\dot{\theta}_2 \Rightarrow -0.366$

$\ddot{\theta}_5 = \dfrac{0.4\times(-0.366)^2 - 0.2\times 22^2\cos(65-167)}{0.3\sin(67-167)}$

$\ddot{\theta}_5 \Rightarrow -68.30.$

$\ddot{r}_7 = \dfrac{-r_6\dot{\theta}_6^2 + r_5\dot{\theta}_5^2\cos(\theta_5-\theta_7) + r_5\ddot{\theta}_5\sin(\theta_5-\theta_7)}{\cos\theta_6}$

where $\dot{\theta}_6 = \dfrac{r_5\dot{\theta}_5\cos\theta_5}{r_6\cos\theta_6}$

$\dfrac{0.5\times(6.81)^2 + 0.3(-68.30)^2\cos(135-155)}{\cos 155}$

$\Rightarrow$ 13.85 m/sec^2

Section 4.2

Comparison of kinematic parameters

In this section I have coincide the four position obtained from graphical method for a complete rotation of crank to the simulated data. In the simulation the complete cycle occurs in 0.6 sec time gap so that I have shown the characteristics up to one complete rotation only the rest curve behaviour is also same as below:

Section 4.2.1

Displacement comparison between simulated and graphical method:

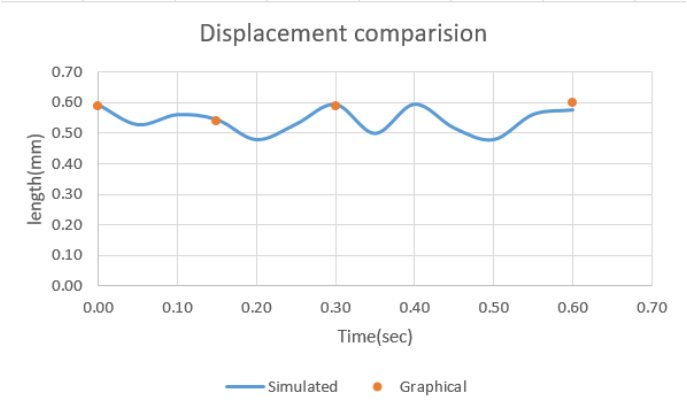


Fig. 4.1 Displacement curve

From above graph the displacement coming from the simulated and graphical are superimposed

Section 4.2.2

Velocity comparison between simulated, Analytical and graphical method:

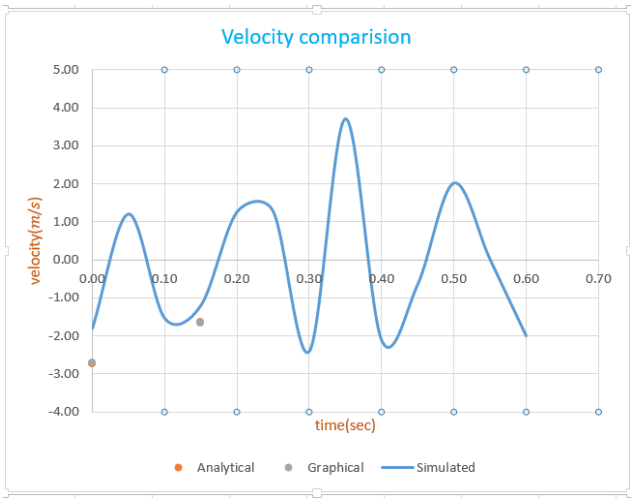


Fig. 4.2 Velocity curve

According to above graph the analytical and graphical result is superimposed but simulated result deviated slightly.

Section 4.2.3

Acceleration comparison between simulated, Analytical and graphical method:

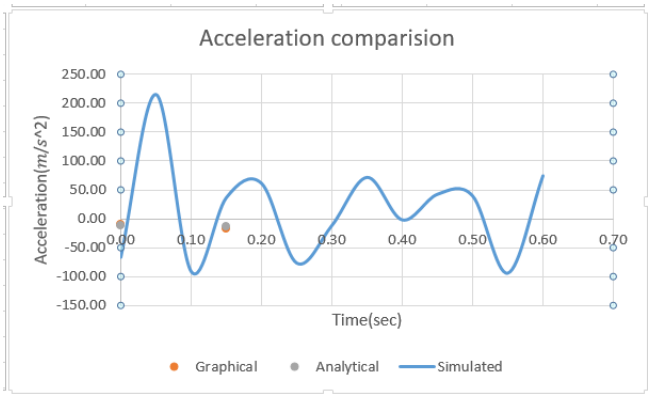


Fig. 4.3 Acceleration curve

At acceleration curve also the same coincidence is occurring where analytical and Graphical result coincide whereas the simulated data curve deviated slightly.

Section 4.2.4-Comment

Position Analysis

In position analysis the simulated result is similar as theoretical and graphical there is less error expected if simulated to higher steps.

Velocity Analysis

Velocity analysis got some error when it comes to coincidence between the simulated and calculated .The result of graphical method and analytical method is exactly matching and coinciding but simulated one is slightly higher. This is may be due to error in convergence during simulation so can be reduced.

Acceleration Analysis

During acceleration analysis all the result are slightly different from each other. During graphical method the result varies due to less calculation precision and Due to small A4 sized paper which may cause geometric error. In simulation the error may be due to less step size. During clearance analysis larger number step gives best convergence with less error so that can be recommended.

References

- *https://cloud.wikis.utexas.edu/wiki/spaces/RMD/pages/51052694/Synthesis+and+prototyping+of+variable+stroke+mechanism*
- *https://youtu.be/PmkLsAqpR5M?si=kWlj8UZjxTiPs3ZT*
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- *https://youtu.be/L61UcSi6aBE?si=zSHGlEnVbPHtdXI8*